

AI CRM Lead Automation

Python 3.11+ local application | Standard-library runtime | Optional OpenAI Responses API | MIT licensed

DELIVERY	ANALYSIS	QUALITY	DEPLOYMENT
Local dashboard + API	Offline or optional OpenAI	69 automated tests	Local-first application

Architecture

INTAKE	ANALYSIS	PERSISTENCE	DELIVERY
Threaded local HTTP server, strict JSON media type, request/message bounds, bearer/session access, CSRF.	Versioned scoring policy or best-effort redacted Structured Outputs; local validation always enforced.	OS interprocess file lock, atomic state, validated legacy cleanup, retention, export/delete.	Escaped dashboard, JSON endpoints, date-accurate metrics, event log, formula-safe CSV.

HTTP surface

METHOD	ROUTE	PURPOSE
GET	/ and /api/health	Dashboard and mode-aware health check
POST	/api/intake	Validate, classify, score, and persist one lead
GET	/api/leads and /api/logs	Structured CRM records, metrics, and audit events
GET/DELETE	/api/leads/{id}	Export or delete one contact record
POST	/api/maintenance/purge	Apply configured contact retention
GET	/export/leads.csv	Formula-neutralized spreadsheet handoff

Security and privacy controls

- Loopback-only host validation	- CSV formula neutralization
- TTL sessions, logout, login throttling	- Legacy PII files removed after migration
- Browser CSRF protection	- Invalid timestamps fail safely
- Security and no-cache headers	- 90-day purge plus export/delete by ID

Verification

69 isolated tests cover configuration, scoring, provider output, retries, interprocess storage, atomic migration, legacy cleanup, retention invariants, session TTL/logout/throttling, CSRF, CSV/HTML safety, and local HTTP behavior. Six-process regression evidence and 91% overall coverage run in CI on Python 3.11-3.13.

Run locally

```
python3 -m unittest discover -s tests -v      python3 src/seed_data.py --reset      python3 src/app.py
```

Verified locally 2026-07-17 | code commit f892334ccf7aebc5eb840ea4b4675f1d55866a72 | github.com/bazhinapolly/ai-crm-lead-automation
No client deployment or measured business outcome claimed. Production rollout requires mandatory identity/authorization, TLS, managed persistence, monitoring, backups, and scheduled lifecycle controls.